Machine Learning: Concepts & Evaluation Metrics

Lecture 1 Knowledge Check & Answer Key | Instructor: Assoc. Prof. Mohamed Abd Elfattah (MET)

Question 1 of 8

What is the primary characteristic of supervised learning?

A) Uses labeled data to learn a mapping from input to output.

B) Learns from unlabeled data to find hidden patterns.

C) Learns via trial and error using rewards.

D) Requires no training data at all.

> **Answer: A** — Supervised learning relies on paired input features (X) and target labels (Y) to learn the mapping function.

Question 2 of 8

Which task involves predicting a continuous numeric value?

A) Classification

B) Regression

C) Clustering

D) Dimensionality reduction

> **Answer: B** — Regression tasks (e.g., house price forecasting) predict continuous numerical outputs.

Question 3 of 8

Which algorithm below is commonly used for clustering?

A) Logistic Regression

B) K-Means

C) Q-Learning

D) Linear Regression

> **Answer: B** — K-Means is a standard unsupervised algorithm used to group unlabelled data into clusters.

Question 4 of 8

In reinforcement learning, what does the agent try to maximize?

A) Prediction accuracy

B) Cumulative reward

C) Number of clusters

D) Size of the training set

> **Answer: B** — The RL agent interacts with an environment to maximize total cumulative rewards through trial and error.

Question 5 of 8

In a confusion matrix, a False Positive occurs when:

A) The model correctly predicts positive.

B) The model correctly predicts negative.

C) The model predicts positive but the actual class is negative.

D) The model predicts negative but the actual class is positive.

> **Answer: C** — False Positive (Type I error) means an incorrect positive prediction when the ground truth is negative.

Question 6 of 8

Which metric matters most when missing a positive case is costly (e.g., disease detection)?

A) Accuracy
B) Precision
C) Recall
D) None of the above

Answer: C — Recall (Sensitivity) measures the proportion of actual positives caught, minimizing False Negatives.

Question 7 of 8

Given TP=90, FN=10, FP=5, TN=895, what is the model's Accuracy?

A) 90.0%
B) 94.7%
C) 98.5%
D) 100%

Answer: C — Accuracy = (TP + TN) / Total = (90 + 895) / 1000 = 985 / 1000 = 98.5%.

Question 8 of 8

Which regression metric penalizes larger errors more heavily, due to squaring?

A) Mean Absolute Error (MAE)
B) Mean Squared Error (MSE)
C) R-squared
D) None of the above

Answer: B — MSE squares the error terms, heavily penalizing large outliers compared to MAE.

Question 9 (Comprehensive Review)

Why is plain Accuracy often misleading when evaluating imbalanced datasets?

A) A model predicting only the majority class achieves high accuracy while missing all minority cases.
B) It only calculates negative instances.
C) It requires perfectly normal distributions.
D) It does not use confusion matrix values.

Answer: A — On a 95/5 split, a lazy model predicting the majority class scores 95% accuracy but 0% recall on the minority class.

Question 10 (Comprehensive Review)

What is the primary purpose of the F1-Score in classification models?

A) To act as the harmonic mean of Precision and Recall, balancing both in a single metric.
B) To calculate total true positives.
C) To measure linear correlation.
D) To compute exact training time.

Answer: A — F1-Score combines Precision and Recall, making it ideal for imbalanced class distributions.